ANALYTICAL METHODS:

1 **DESCRIPTION**

1.1 Physically examine a representative sample of the material for Appearance and other characteristics: it is **A** white flake, crystalline powder, hygroscopic, freely soluble in water, soluble in alcohol**.**

2 **IDENTIFICATION**

2.1 Calcium Test

2.1.1 Accurately weigh and transfer about 10g of sample into a 100ml volumetric flask.

2.1.2 Add 60ml of distilled water and mechanically shake/sonicate till dissolution is complete.

2.1.3 Dilute to the mark with distilled water and mix.

2.1.4 Pipette 1ml of sample solution into a test tube.

2.1.5 Add 5ml of 5M acetic acid, 0.5ml of 5.3%w/v solution of potassium ferrocyanide (potassium hexacyanoferrate II) and mix: Solution remains clear

2.1.6 Add about 50 mg of ammonium chloride and mix: A white, crystalline precipitate is produced.

2.2 Chloride Test

Reagents

2.2.1 2M nitric acid

Pipette 12.6ml of concentrated Nitric acid to 100ml volumetric flask containing 60ml of distilled water , dilute to the mark with distilled water and mix

2.2.2 4.25% w/v silver nitrate

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Accurately weigh and transfer about 4.25g silver nitrate to 100ml volumetric flask, dilute to the mark with distilled water and mix

2.2.3 10M ammonia

Dilute about 41g of con ammonia to 100ml volumetric flask, dilute to the mark with distilled water and mix

2.2.4 Sample preparation

2.2.5 Accurately weigh and transfer about 1g of sample into a test tube

2.2.6 Add 2 ml of distilled water and shake until dissolution is complete.

2.2.7 Acidify with 2M nitric acid, add 0.4 ml of 4.25% w/v silver nitrate solution.

2.2.8 Shake and allow to stand; **a** curdy, white precipitate is produced

2.2.9 Filter and wash the precipitate with three 1-ml quantities of water.

2.2.10 Carry out this operation rapidly in subdued light, disregarding the fact that the supernatant solution may not become perfectly clear: Suspend the precipitate in 2 ml of water and add 1.5ml of 10M ammonia; the precipitate dissolves easily with the possible exception of a few large particles which dissolve slowly**.**

3 ACIDITY OR ALKALINITY

3.1 Reagents

3.1.1 0.01M hydrochloric acid; dilute 10ml of 0.01M hydrochloric acid to 100ml with distilled water

3.1.2 0.01M sodium hydroxide ; dilute 10ml of 0.1M sodium hydroxide to 100ml with distilled water

3.1.3 1%w/v solution of phenolphthalein in ethanol; Accurately weigh and transfer about 1.0g to 100ml volumetric flask, dilute to the mark with ethanol and mix

3.2 Procedure

3.2.1 Accurately weigh and transfer about 10g of sample into a 100ml volumetric flask.

3.2.2 Add 60ml of distilled water and mechanically shake/sonicate till dissolution is complete.

3.2.3 Dilute to the mark with same solvent and mix.

3.2.4 Pipette 10.0ml of this solution into a 100ml conical flask, add 0.5ml of 0.1%w/v phenolphthalein in ethanol.

3.2.5 Titrate with 0.01M hydrochloric acid to the disappearance of pinkish colouration or with sodium hydroxide solution, the appearance of pinkish colouration: Not more than 0.2ml of 0.01M hydrochloric acid or 0.01M sodium hydroxide is required to change the colour of the indicator.

4 SULPHATES.

4.1 Sulphate Standard Preparation 1000ppm

4.1.1 Accurately weigh and transfer about 0.543g of analytic grade potassium sulphate into a 100ml volumetric flask.

4.1.2 Add 60ml of 30% v/v ethanol in distilled water and sonicate till dissolution is complete.

4.1.3 Dilute to the mark with 30% v/v ethanol in water and mix “(1000ppm sulphate standard stock solution).

4.1.4 Pipette 15.0ml of this solution into another 100ml volumetric flask, dilute to the mark with 30% v/v ethanol in water and mix (150ppm sulphate working standard).

4.2 Sample Preparation

4.2.1 Accurately weigh and transfer about 10g of sample into a 100ml volumetric flask.

4.2.2 Add 60ml of distilled water and mechanically shake/sonicate till dissolution is complete.

4.2.3 Dilute to the mark with same solvent and mix.

4.3 Procedure

4.3.1 Pipette 5.0ml each of this solution and 1ml of acetic acid into 25ml volumetric flasks.

4.3.2 To another flask, 13.0ml of distilled water, 2ml of 300ppm sulphate standard solution, and 1ml of acetic acid into 25ml volumetric flasks.

4.3.3 To each flask, add 1ml of 25% w/v solution of barium chloride solution and 1ml of 2M nitric acid.

4.3.4 Dilute to the mark with 30% v/v ethanol in water and mix.

4.3.5 Shake each flask and allow standing for 5 minutes: Any opalescence produced is not more intense than that of Sulphate standard (300ppm SO_4).

5 **BARIUM**

5.1 Calcium Sulphate Solution

5.1.1 Accurately weigh and transfer about 5g of analytic grade calcium sulphate into a 100ml volumetric flask.

5.1.2 Add 60ml of distilled water and mechanically shake/sonicate for 1 hour.

5.1.3 Dilute to the mark with distilled water and mix.

5.1.4 Filter through Whatman #1 filter paper into another 100ml volumetric flask.

5.2 Sample Preparation

5.2.1 Accurately weigh and transfer about 10g of sample into a 100ml volumetric flask.

5.2.2 Add 60ml of distilled water and mechanically shake/sonicate till dissolution is complete.

5.2.3 Dilute to the mark with same solvent and mix.

5.3 Procedure

5.3.1 Pipette 10.0ml of this solution and 1ml of calcium sulphate solution into a test tube and mix:

5.3.2 Pipette 10.0ml of this solution and 1ml of distilled water into another test tube and mix:

5.3.3 Allow 15 minutes and compare the turbidity of the two solutions: After at least 15 minutes, any opalescence in the sample solution is not more intense than that in a mixture of 10 ml of solution S and 1 ml of distilled water

6 **HEAVY METALS**

6.1 Lead Stock Solution(100ppm)

6.1.1 Accurately weigh and transfer about 160mg of Lead into a 1-litre volumetric flask.

6.1.2 Add 200ml of distilled water, 1ml of concentrated nitric acid, cover and shake till dissolution is complete.

6.1.3 Dilute with distilled water to 1000ml and mix. Prepare and store this solution in glass containers free from soluble lead salts.

6.2 Lead stock solution (1ppm

6.2.1 Pipette1.0ml of the 100ppm lead standard solution into a 100ml volumetric, dilute to the mark with distilled water and mix.

6.3 Buffer Solution pH3.5

6.3.1 In a 250ml beaker, dissolve 25.0g of ammonium acetate in 25ml of water and add 38.0ml of 7M hydrochloric acid.

6.3.2 Adjust the pH if necessary with 2M hydrochloric acid or 6M ammonium hydroxide and dilute to 100.0ml with water.

6.4 Thioacetamide SOLUTION(4%)

6.4.1 Dissolve 4g of thioacetamide in a 100ml volumetric flask, make up with distilled water

Thioacetamide Reagent

To 2ml of thioacetamide solution in a 100ml volumetric flask, add 7.5ml of water, 37.5ml of M sodium hydroxide and 50ml of glycerol (85%). Heat in a water-bath for 20 seconds. Prepare immediately before use.

6.5 Sample Preparation

6.5.1 Accurately weigh and transfer about 10g of sample into a 100ml volumetric flask.

6.5.2 Add 60ml of distilled water and mechanically shake/sonicate till dissolution is complete.

6.5.3 Dilute to the mark with same solvent and mix.

6.5.4 To 12ml of the solution thus obtained (in a test tube), add 2ml of buffer solution pH3.5, 1.2ml of thioacetamide reagent and immediately mix.

6.6 Standard Preparation

6.6.1 Transfer 10 ml of 1ppm *lead standard solution* and 2 ml of the sample solution into a suitable test tube.

6.6.2 To the solution thus obtained, add 2ml of buffer solution pH3.5, 1.2ml of thioacetamide reagent and immediately mix.

6.7 <u>Blank Preparation</u>

6.7.1 Prepare a blank using a mixture of 10ml of water and 2ml of sample solution

6.7.2 To the solution thus obtained, add 2ml of buffer solution pH3.5, 1.2ml of thioacetamide reagent and immediately mix.

6.8 OBSERVATION

6.8.1 Compared to the blank, after 2 minutes any brown colour in the solution obtained with the sample is not more intense than in the standard solution

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7 **Appearance of solution**

7.1 Dissolve 10.0 g of sample in *carbon dioxide-free waterR*

7.2 and dilute to 100 mL with the same solvent

7.3 SolutionS is clear and not more intensely coloured than reference solution Y.

8 **IRON**

8.1 10ppm Iron Standard Stock

8.1.1 Accurately weigh and transfer about 70.2mg of Ferrous Ammonium Sulphate hexahydrate [$Fe(NH_4)_2(SO_4)_6H_2O$] into a one-litre volumetric flask.

8.1.2 Add 300ml of distilled water, 25ml of concentrated sulphuric acid, mechanically shake/sonicate till dissolution is complete.

8.1.3 Dilute to the mark with distilled water and mix. KEEP THIS SOLUTION FOR NOT MORE THAN 3 MONTHS!

8.2 Sample Preparation

8.2.1 Accurately weigh and transfer about 10g of sample into a 100ml volumetric flask.

8.2.2 Add 60ml of distilled water and mechanically shake/sonicate till dissolution is complete.

8.2.3 Dilute to the mark with same solvent and mix.

8.3 Procedure

8.3.1 Pipette 10.0ml each of the sample and standard preparations into separate 25ml volumetric flask.

8.3.2 Add 2ml of 20%w/v citric acid, 0.1ml of mercpatoacetic acid, mix and compare sample and

standard preparations

8.3.3 Dilute to the mark with distilled water and mix.

8.3.4 Using a suitable measuring cylinder, transfer 20ml of each preparation into separate Nessler Cylinders and compare the intensity of the colour developed: Pink colouration obtained in the test solution is not more intense than the standard solution.

9 **ASSAY**

9.1 Method A

9.11 0.1M Disodium Edetate (Preparation and Standardization)

Procedure

9.1.2 Accurately weigh and transfer about 280mg each into two separate 500ml conical flasks.

9.1.3 Add 300 ml of distilled water and sonicate/swirl till dissolution is complete.

9.1.4 Add 15mg of calconcarboxylic acid triturate and titrate with 0.1M Disodium Edetate VS until the colour changes from violet to full blue

9.1.5 Calculate the assay as follows:

$$\%\text{Assay} = \frac{\text{Titre} \times 14.70\text{mg} \times \text{standard factor of EDTA} \times 100}{\text{Sample weight (mg)}}$$

9.2 Method B

Reagents

9.2.1 Ammonia Buffer pH 10.0

Weigh about 5.4 of ammonium chloride into 100ml volumetric flask.

Add 20ml of distilled water, cover and shake till dissolution is complete.

Add 35ml of 10M ammonia, dilute to the mark with distilled water and mix

9.2.2 Mordant Black II Ttriturate

Weigh about 1g of mordant black II into a suitable mortar
Weigh and add 99g of sodium chloride to the same mortar
Mix the powder together, then grind into a very fine powder. Store in a desiccators.

9.2.3 0.1M Disodium Edetate VS prepared

Procedure

9.2.4 Accurately weigh about 280mg each into two separate 100ml conical flasks.
9.2.2 Add 100ml of distilled water and sonicate till dissolution is complete
9.2.3 Add in the following order with continuous mixing: 5ml of ammonia buffer pH 10.0 and 50mg of mordant black II triturate to each flask and mix.
9.2.4 Titrate with 0.1M disodium edetate from violet to a bluish end-point.
9.2.5

$$\%Assay = \frac{Titre \times 14.70mg \times Standard\ Factor\ of\ EDTA \times 100}{Sample\ weight\ (mg)}$$

10 MAGNESIUM AND ALKALI METALS

10.1 Sample Preparation

10.1.1 Accurately weigh and transfer about 10g of sample into a 100ml volumetric flask.
10.1.2 Add 60ml of distilled water and mechanically shake/sonicate till dissolution is complete.
10.1.3 Dilute to the mark with same solvent and mix.
10.1.4 Pipette 20ml of this solution into 100ml beaker.
10.1.5 Add 80ml of distilled water, 2g of ammonium chloride, 2ml of 50%v/v ammonia and mix.
10.1.6 Transfer onto hot plate and heat to boiling

10.1.7 Pour into a 250ml beaker containing 75ml of 7%w/v solution of ammonium oxalate
10.1.8 Allow to standing for 4 hours and dilute to 200ml with distilled water.
10.1.9 Filter through Whatman filter paper into another 250ml beaker.
10.1.10 Pipette 100ml of the filtrate into another, add 0.5ml of concentrated sulphuric acid and mix.
10.1.11 Transfer beaker onto a hotplate and evaporate to 3ml
10.1.12 Transfer the residue into a pre-weighed porcelain crucible (w1).
10.1.13 Rinse the beaker with 3 x 1ml of distilled water and transfer into the same crucible.
10.1.14 Transfer crucible onto a boiling water bath and evaporate to dryness.
10.1.15 Transfer to a furnace operating at 600°C and for about 1 hour.
10.1.16 Transfer into a desiccator and cool allow cooling to room temperature and re-weigh (W2).
10.1.17 Calculate the percentage residue as follows:

$$\text{Magnesium \& Alkaline Metals} = \frac{(W2-W1) \times 100 \times 100}{\text{Sample Weight (g)} \times 20}$$

11 ALUMINUM

11.1 Acetate Buffer PH 6.0
11.1.1 Weigh about 40g of ammonium acetate into a 250ml beaker
11.1.2 Add 100ml of distilled water, 1.6ml of glacial acetic acid and mix till dissolution is complete.
11.1.3 Adjust the pH to 6.0, if necessary, using 10M ammonia or 5M acetic acid
11.1.4 Dilute to the 200ml- mark with distilled water and mix.

11.2 200ppm Aluminium Standard Solution
11.2.1 Weigh about 0.352g of [Aluminium Potassium Sulphate ($AlK(SO_4)_2,12H_2O$) into a 100ml volumetric flask.

11.2.2 Add 30ml of distilled water, 10 ml of 1M sulphuric acid and mechanically shake/sonicate till dissolution is complete.

11.2.3 Dilute to the 100ml- mark with distilled water and mix. Store this solution for not more than 3 months.

11.3 2ppm Aluminium Standard Solution (Prepare Immediately Before Use**!)**

11.3.1 Pipette 1.0ml of 200ppm aluminium solution into 100ml volumetric flask.

11.3.2 Dilute to the 100ml- mark with distilled water and mix. Store this so

11.4 0.5% w/v Solution Of 8-Hydroxyquinoline

11.4.1 Weigh and transfer about 1g of 8-hydroxyquinoline into a dry 200ml volumetric flask.

11.4.2 Add 100ml of chloroform, mechanically shake/sonicate till dissolution is complete.

11.4.3 Dilute to the 200ml- mark with chloroform and mix.

11.5 Sample Preparation

11.5.1 Accurately weigh and transfer about 4g of sample into a 100ml volumetric flask.

11.5.2 Add 60ml of distilled water and mechanically shake/sonicate till dissolution is complete.

11.5.3 Add 10ml of acetate buffer pH 6.0, dilute to the mark with distilled water and mix.

11.5.4 Pipette 20ml of this solution into 100ml beaker.

11.6 Procedure

11.6.1 Into three 250 ml Separate funnels, pipette and add in the following order:

11.6.2

Reagent s	Sample	Standard	Blank
Sample Solution	100.0ml	-	-
2ppm Aluminium Standard Solution	-	2.0ml	-
Water	-	98.0ml	100.0ml
Acetate Buffer pH 6.0	10.0ml	10.0ml	10.0ml

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11.6.3 Separate extract each of the sample, standard and blank preparations in the above table with successive quantities of 20, 20 and 10 ml of a 0.5% w/v solution of 8-hydroxyquinoline in chloroform, filtering the extracts through chloroform-wet Whatman #1 filter paper into separate, 'sample', 'standard', 'blank' labeled, dry 50ml volumetric flasks

12 Microbiological Tests

12.1 <u>Limulus Amebocyte Lysate LAL Test</u>

12.1.1 Open the seals of the test tubes of lyophilized LAL Reagents

12.1.2 Add 0.2ml LAL Reagent water, endotoxin standards or test sample into each tubes.

12.1.3 Swirl gently but thoroughly to rehydrate the LAL reagent. Do not shake or vortex as content will foam.

NOTE: cover the reaction tubes with pyrogen free aluminium foil

12.1.4 Incubate the reaction mixture immediately at 37°c for 60minutes. Avoiding vibration during incubation.

12.1.5 If large numbers of samples are tested in parallel, the tests should be batched and started at intervals that permit the reading of each within the time limit.

12.1.6 Sample to be tested should be stored under conditions that bacteriological activities are controlled

12.1.7 Read reaction tubes one by one.

12.1.8 Invert the tube in one smooth motion.

12.1.9 A positive result is noticed by the formation of a gel which does not collapse when the tube is inverted. A negative result is characterized by the absence of solid clot after inversion.

Testing Methodology of Bacterial Endotoxin (LAL)

12.2.1 Each specimen shall be tested in duplicate

12.2.2 NMC: Take 100µl(0.1ml) LRW

12.2.3 Take 50µl (0.05ml) LRW and 50µl (0.05ml) test dilution.

12.2.4 Take 50µl(0.05ml) test dilution and 50µl(0.05ml) CSE of 4λ

12.2.5 Add the LAL reagents as mentioned below in the table 1

12.2.6 Mix gently and incubate the tubes in heating block at 37^0C ± 1 for 60min.

12.2.7 After incubation, remove the tubes gently from the heating block and slowly invert through 180^0C and note the observations.

12.2.8 If a firm gel has formed, record the results as positive.

12.2.9 A result is negative, if an intact gel is not formed.

12.2.10 Record the results and other details as per format no. "Analytical Test Data Sheet for Bacterial Endotoxin (LAL) Test" in case of sterile raw material.

12.3	**Pyrogen** 12.3.1 Perform the test in a separate area designated for Pyrogen testing and under environmental conditions similar to those under which the rabbits are housed. 12.3.2 Withhold all food from the rabbits used during the period of the test 12.3.3 Draw out the required number of rabbits from their living room according to the number of tests planned. Always select more rabbits than actually required to counter any exclusion of animals in the initial measuring. 12.3.4 All the rabbits which are to be used for the test should be more than 1.5kg weight, should be healthy and must not have been used for any test in previous three(3) days and in previous three (3)weeks if any test failed using that rabbit. 12.3.5 Put the selected rabbits to testing cages. Take the initial temperature of each rabbit by inserting the thermometer to in the anus rectum of the rabbit to a depth of about 7.5cm. 12.3.6 If the initial temperature is less than 38.8^{0}c or more than 39.8^{o}cexclude the animal from the test. 12.3.7 Ensure all the needles and syringe to be used are new 12.3.8 Arrange the rabbits into groups of three taking into consideration that the initial temperature of the rabbits in a group should not differ by more than 1^{0}c.

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12.3.9 Take another two reading of temperature in 30 min intervals between 1 and 3 hours subsequent to the injection.

12.3.10 Any rabbits showing more than 0.2^0c temperature difference between two successive readings in the last three readings can be excluded from the test.

12.3.11 Calculate the initial temperatures of the animals by taking the average of the temperature readings taken in the last 40 minutes.

12.3.12 Inject the solution intended to be tested to the marginal veins of the rabbits of a group slowly but not exceeding 10 minutes after the start of administration. The quantity of test solution depends upon the body weight of the rabbits and the test solution.

12.3.13 If the substance to be tested is not in aqueous sterilized solution form, consult the official monograph regarding the preparation of simulating solution and the volume of the preparation to be injected.

12.3.14 After injecting the solution, Record the temperature at 30 minutes intervals between 1 and 3 hours subsequent to the injection.

12.3.15 The test involves measuring the rise in temperature of rabbits following the intravenous injection of a test solution.

12.4 **MLT**

12.4.1 Weigh 1g or 1ml of the raw material sample and dissolve aseptically in 9ml sterile soybean casein digest medium prepared as per SOP for media preparation (BML-QDG-MC-011).

12.4.2 Determine the Total Aerobic Microbial count (TAMC) and Total Yeast and Molds count (TYMC) by using pour plate method.

12.4.3 Testing to be performed under LAF of specified area

12.4.4 Arrange all the required material on the LAF before starting the test.

12.4.5 Aseptically Pipette 1ml of the solution from above in a sterile Petri dishes

12.4.6 Add approximately 15-20ml of sterile Soyabean Casein Digest Agar prepared as per SOP(BML-QDG-MC-011), (cooled approximately up to 45°C) in two petri plates for TAMC

12.4.7 Add approximately 15-20ml of sterile Saboraud Dextrose Agar prepared as per SOP (cooled approximately up to 45°C) in two petri- plates for TYMC

12.4.8 Gently swirl the contents inside the petri-dishes for proper mixing of the sample, allow medium to solidify, taking the precaution of not spilling the media from the plate

12.4.9 Simultaneously perform the negative control by plating 1 ml of diluents used instead of the sample for each medium. Add 15-20ml of the media previously maintained not more than 45°c and allow for solidification.

12.4.10 Incubate the plates at 30°C to 35°C for 3 - 5 days for soybean Casein Digest Agar for TAMC and plates of Saboraud Dextrose Agar at 20^{0}C to 25°C for 5 days in an inverted position.

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